

Ocular Tumors

Visual Outcomes of Parapapillary Uveal Melanomas Following Proton Beam Therapy



Juliette Thariat, MD, PhD,* Jean-Daniel Grange, MD,[†] Carlo Mosci, MD,[‡]
Laurence Rosier, MD,[§] Celia Maschi, MD,^{||} Francesco Lanza, MD,[‡]
Anh Minh Nguyen, MD,[†] Franck Jaspart, MD,[¶] Franck Bacin, MD, PhD,[¶]
Nicolas Bonnin, MD,[¶] David Gaucher, MD,[#]
Wolfgang Sauerwein, MD, PhD,** Gaelle Angellier, PhD,*
Joel Hérault, PhD,* and Jean-Pierre Caujolle, MD^{||}

*Department of Radiation Therapy, Cancer Center Antoine Lacassagne-Nice Sophia Antipolis University Hospital, Nice, France; [†]Department of Ophthalmology, Eye University Clinic la Croix Rousse, Lyon, France; [‡]Department of Ophthalmology, National Institute for Cancer Research, Mura Delle Cappucine, Genova, Italy; [§]Eye Clinic, Centre d'Exploration et de Traitement de la Retine et de la Macula, Bordeaux, France; ^{||}Department of Ophthalmology, Eye University Clinic Pasteur 2, Nice, France; [¶]Department of Ophthalmology, Eye University Clinic Gabriel Montpied, Clermont Ferrand, France; [#]Department of Ophthalmology, Eye University Clinic, Hopital Civil, Strasbourg, France; and **Department of Radiation Therapy, NCTeam, Strahlenklinik, Universitätsklinikum Essen, Essen, Germany

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Summary

Parapapillary melanoma patients have been treated conservatively with proton therapy for 25 years. Patient, tumor, initial visual acuity, treatment, and radiation-induced complications influence visual acuity. However, a personalized computed tomography-based approach using a wedge filter to limit irradiation of the papilla and optic nerve length or macula

Purpose: In parapapillary melanoma patients, radiation-induced optic complications are frequent and visual acuity is often compromised. We investigated dose-effect relationships for the optic nerve with respect to visual acuity after proton therapy.

Methods and Materials: Of 5205 patients treated between 1991 and 2014, those treated using computed tomography (CT)-based planning to 52 Gy (prescribed dose, not accounting for relative biologic effectiveness correction of 1.1) in 4 fractions, with minimal 6-month follow-up and documented initial and last visual acuity, were included. Deterioration of ≥ 0.3 logMAR between initial and last visual acuity results was reported.

Results: A total of 865 consecutive patients were included. Median follow-up was 69 months, mean age was 61.7 years, tumor abutted the papilla in 35.1% of patients, and tumor-to-fovea distance was ≤ 3 mm in 74.2% of patients. Five-year relapse-free survival rate was 92.7%. Visual acuity was $\geq 20/200$ in 72.6% of patients initially and 47.2% at last follow-up. A wedge filter was used in 47.8% of the patients, with a

Reprint requests to: Juliette Thariat, MD, PhD, Department of Radiation Therapy, Centre Lacassagne, 227 Ave de la Lanterne, 06200 Nice, France. Tel: 33 (0)492031083; E-mail: jthariat@gmail.com

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J.T., J.H., and J-P.C. contributed equally to the manuscript.

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